Malqaf L5 · the courtyard’s north-west corner · a decision-grade study in phases

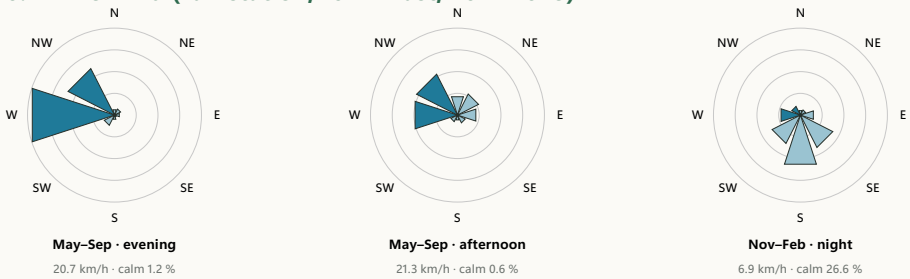
STUDY ONLY: no design file is changed until the owner adopts L5 · the owner’s decisions D1–D11 (9 Nov 2026) are the brief · **PHASE 0: the geometry check and the climate basis** (the next phases follow the owner’s reply) · every figure **ASSUMED** unless measured; final sizes by the structural and MEP engineers.

0.1 · The brief’s coordinates against the repo (V9.3)

The brief	The repo	
Courtyard x 4.25–8.95, y 6.5–13.0	rooms_gf.csv: 4.7 × 6.5 centred 6.6 / 9.75	✓
Seating zone x 5.925–8.85, y 7.1–12.3	air_system.typ `air-sunken`	✓
GF covered strip x 4.25–5.80, y 6.5–13.0	the FF projection 1.55 m (B4 x 4.25 → B3 x 5.80); clear ≈ 1.35–1.45 (the riwaq’s glazing line and the colonnade)	✓
Women’s majlis x 4.25–8.95, y 0–6.5 · FF family living x 4.25–8.95, y 3.0–6.5	rooms_gf.csv, rooms_ff.csv	✓
Family lobby + riwaq x 2.25–4.25	**partly:** x 2.25–4.25 only for y 0–2.2 (the lobby); **along the courtyard x 2.75–4.25 (1.5 m)**	≠
Riwaq glazing on x 4.25	glass y 6.8–8.9 and 10.7–12.7, the courtyard glass door y 9.1–10.5 (presentation.typ)	✓
FF gallery x 4.25–5.80 (y 6.5–19.25) + x 2.75–4.25 (y 2.2–9.2), ≈ 3.05 m wide at y 6.5–9.2	rooms_ff.csv (also x 2.25–4.25 for y 0.7–2.2)	✓
W1 × 6 on x 5.80: y 6.783–8.383 … 17.408–19.008; sill +3.82, springing +5.80, crown +6.60	ff_family.typ, 3d/params.yaml	✓
C21 (5.8, 6.5), C22 (5.8, 13.0), C23 (5.8, 19.25), Ø 0.30 · and C17 at (4.25, 6.5)	structure.typ (C17: 0.25 × 0.50, to the roof)	✓
B3 on x 5.80, 6.50 + 6.25, upturned or drop · B4 on x 4.25, drop, soffit +3.10 · R11 on C21–C22, soffit +6.75 · the projection slab 0.20 E–W 1.55 m	structure.typ	✓
Levels GF ±0.00, FF +3.60, roof finish +7.30, attic coping +8.68 · Z3 x 2.85–4.15, y 1.0–3.5	roof.typ (V9.2 attic)	✓
Baseline L1 (V9.3): NE corner, shaft 0.90 m², crown +12.18	air_system.typ, roof.typ (V9.3)	✓

Mismatches: (1) the riwaq is 1.5 m wide along the courtyard (x 2.75–4.25), 2.0 m only at the lobby (y 0–2.2); (2) the covered strip’s clear width ≈ 1.35–1.45 m depending on the glazing build-up (to fix in Phase 1). **Also noted:** a column C17 already stands at (4.25, 6.5), the corner the L5 core starts from (Phase 2); at y 6.5–9.2 the FF gallery is 3.05 m wide only because its west half (x 2.75–4.25) is the stair’s arrival zone: a core across the whole strip (V-b) leaves 1.5 m on axes, ≈ 1.3 clear, **below the brief’s 1.5 m clear** (⚠ caution for Phase 1).

0.2 · The wind (Taif station, 10 m mast, 2011–2025)



Mon	N	NE	E	SE	S	SW	W	NW	km/h	calm
Jan	7.1	5.9	10.1	12.9	21	16.1	16.3	10.6	11.7	14.2
Feb	6.3	6.3	11.3	12.4	17.9	18.2	17.7	9.8	13.9	10.9
Mar	4.2	4.1	7.8	11	19.1	17.4	24.9	11.4	15	8.5
Apr	3.2	4	7.1	11.3	18.2	19.4	26.5	10.2	14.9	6.7
May	4.5	5.2	6.2	7.1	13	16.9	33.2	13.9	13.8	6.8
Jun	8	8.4	6.4	3.9	4.4	8	35.7	25.3	16.8	3.8
Jul	7.2	3.8	1.4	0.6	1.1	4.2	44	37.6	21.7	1.4
Aug	8.9	6.2	2.9	1.4	2.1	5	40.7	32.8	19.6	2.3
Sep	7.2	10.6	10.8	7.2	8	10.6	25	20.7	13.8	7.4
Oct	3.4	6.8	14.3	19.1	18.9	10.8	15.9	10.7	11.5	12.1
Nov	4.6	4.5	10.3	15.2	22.1	14.1	17.6	11.7	10.7	13.9
Dec	5.6	4.7	12	15.8	21.2	14.1	15.1	11.3	10.9	14.5

% of the non-calm hours by the direction the wind blows FROM; calm < 2 km/h. Bold blue: ≥ 25 %. Station at the airport (21.48 N 40.54 E, 1 478 m), **ASSUMED representative of the farm** (its terrain and obstacles to be checked on site).

Phase 0 · when the malqaf works · dust · summer cooling · winter closing

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Tmax / Tmin (°C)	23 / 10.2	25 / 11.7	27.6 / 14.6	30 / 17.2	33.5 / 20.4	36.2 / 24	35.5 / 25.3	35.5 / 24.9	35 / 22.6	31 / 17.9	26.6 / 14.1	23.9 / 11.3
the coldest nights (p10, °C)	7	8	11	14	18	21	23	23	20	15	12	8
15 h: T · RH	22.2 · 36%	24.3 · 30%	26.9 · 25%	28.9 · 26%	32.4 · 19%	35.6 · 12%	34.9 · 15%	34.7 · 20%	34.2 · 15%	30.2 · 19%	25.6 · 34%	23.3 · 36%
21 h: T · RH	16.3 · 57%	18.2 · 50%	20.8 · 45%	22.7 · 46%	26.2 · 36%	29.3 · 24%	29 · 26%	28.5 · 37%	28 · 35%	24 · 45%	19.8 · 59%	17.3 · 57%
evening wind (km/h)	12	14.7	16.4	16.9	17.5	21.1	24.6	22.9	17.5	13.3	11	11
dust days / year	3.1	6.3	11	8.3	5	8	5.5	7.8	3.7	1.2	0.5	1.8
seating 16–23: dust hours	4.7 %	8.5 %	17.3 %	10.8 %	6.4 %	12.1 %	8.4 %	11.4 %	4.8 %	1.3 %	0.7 %	2.9 %
pleasant & dust-free	**21 %**	**43 %**	**65 %**	**70 %**	**76 %**	**47 %**	**53 %**	**62 %**	**71 %**	**89 %**	**65 %**	**34 %**
≈ hours / month	53	96	162	169	188	113	132	154	170	221	157	84
too hot > 32 °C	0 %	0 %	0 %	1 %	14 %	36 %	29 %	25 %	22 %	2 %	0 %	0 %
too cool < 18 °C	46 %	24 %	8 %	5 %	1 %	0 %	0 %	0 %	0 %	1 %	11 %	35 %
cooler outlet at 15 h (°C)	–	–	–	18.6	19.8	20.3	20.8	21.8	20.2	18.3	–	–

Pleasant = 20–32 °C with 7–35 km/h at the 10 m mast (the head at ≈ +12 m: ≈ 4 % more), the family's seating hours 16:00–23:00 (**ASSUMED**); dust = an ISD synoptic report with dust or sand in the present weather (06–09, 30–35). Cooler = a direct evaporative cooler at 80 % saturation efficiency on the 15 h mean (wet bulb by Stull 2011), **ASSUMED**.

0.3 · What the climate says

- **When the malqaf works: ≈ 1699 h a year** of pleasant, dust-free seating hours (58 % of 16:00–23:00): the best months **October (89 %)**, May (76 %), April, September, March and November (65–71 %); the summer evenings 47–62 % (the rest too hot → the cooler, D9); December–January 21–34 % (too cool → closed).
- **The faces that matter:** the warm-season evening wind comes **from the west (≈ 50 %) and the north-west (≈ 30 %)**, at ≈ 21 km/h (≈ 6 m/s, calm 1 %); the afternoons add the north and north-east (≈ 25 %). **Priority: the W face, then the N face** (a NW wind strikes both at 45°); E and S faces matter only in the cool season's light night winds (S / SE, calm 27 %), when the malqaf is closed anyway.
- **Dust: ≈ 62 dust days a year** (mostly dust in suspension; a dust-storm code or a visibility below 1 km on less than 1 day a year), the most in **March (11)** and June–August (8); ≈ 5–17 % of the seating hours. The dust arrives as often from the east and south-east (≈ 35 %) as from the west: **a dust trap at the foot and a PM2.5 sensor are essential**; the filtered path must never depend on the breeze path.
- **Summer cooling (D9):** the afternoon air is very dry (RH 12–20 %, the wet bulb 16–19 °C): a direct evaporative cooler would deliver **≈ 20–22 °C air** from 35 °C, June–September: an excellent potential for a mobile cooler in the shaft's foot.
- **Winter:** the evenings below 18 °C on ≈ 35–46 % of December–January seating hours, the coldest nights ≈ 7–8 °C: the breeze path must close tight and insulated (D5).
- ⚠️ **CAUTION** · the station is the airport's 10 m mast; the farm's own exposure (terrain, trees, the old house to the west) is unknown: a simple anemometer on the roof for a few weeks would confirm the faces before Phase 5.

0.4 · Thresholds for the control modes (to finalise in Phase 4)

Mode	Switching (outdoor sensors; **ASSUMED**)
Normal	breeze damper open: 20 ≤ T ≤ 32 °C and PM2.5 ≤ 35 µg/m³; the filtered fan always on
Clear day	also the gallery window: 18 ≤ T ≤ 30 °C, PM2.5 ≤ 15, wind ≤ 30 km/h
Dust	all breeze openings shut: PM2.5 > 50 for 10 min (reopen < 25 for 30 min); the fan up for positive pressure
Cold	breeze shut: T < 18 °C (or < 20 °C after sunset, Nov–Mar); the fan at its minimum
Summer	T > 32 °C: the cooler on in the foot, the breeze damper open to feed it, the gallery window shut
High wind	gusts > 45 km/h: the damper half-closed (dust and noise)

Open questions for the owner

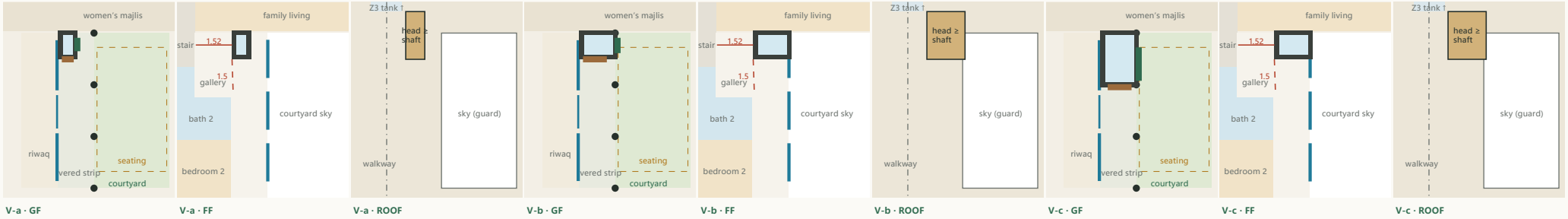
1. Where exactly is the farm (distance and height from the airport; hills or trees to the west and north-west)? Can a small roof anemometer run for a few weeks?
2. Are 16:00–23:00 and 20–32 °C the right seating hours and comfort band for the family?
3. Accept the PM2.5 thresholds (35 / 50 µg/m³) and the winter closing at 18 °C?
4. L5 frees the north-east corner: the corner study's bedroom swap (`studies/ne_corner.typ`) is then not needed. Confirm Phase 1 can assume the corner returns to its V9.1 rooms?

ملخص المرحلة 0: الرياح في الطائف صيفا مساءً غربية (نحو 50٪) وشمالية غربية (نحو 30٪) بسرعة نحو 21 كم/س، فالأولوية لوجهي البرج الغربي ثم الشمالي. في السنة نحو 1699 ساعة جلوس (من 4 عصرا إلى 11 ليلا) لطيفة وخالية من الغبار، وأفضلها أكتوبر ومايو والربيع والخريف؛ صيفا تكون الحرارة عالية في ثلث الوقت تقريبا فيعمل المبرد الصحراوي (الهواء جاف جدا فيخرج المبرد هواءً بنحو 20–22°م)، وشتاءً يُغلق الملقف حين تقل الحرارة عن 18°م. الغبار نحو 62 يوما في السنة وأكثره في مارس والصيف، فلا بد من مصيدة غبار وحساس للغبار. البيانات من محطة مطار الطائف وتحتاج تأكيدا في موقع المزرعة.

Phase 0 checks **passed** (4) · **Sources:** Meteostat bulk hourly data, station 41036 Taif (bulk.meteostat.net), 2011–2025, 128 500 records; NOAA NCEI Integrated Surface Database (ISD) global-hourly, station 41036099999, 2015–2024, synoptic reports (noaa-global-hourly-pds); Windfinder, Taif Regional Airport statistics (prevailing W, 259°, 2011–2026); weatherandclimate.eu, Climate of Taif (monthly normals, which agree with the station means above); "Spatial and temporal variations in the incidence of dust storms in Saudi Arabia revealed from in situ observations" (Geosciences 2019; Cranfield and Exeter repositories: the west 5–15 dust-storm events a year, the spring peak). Reproduce: `python tools/climate_taif.py`.

Phase 1 · L5 geometry and variants · the north-west corner of the courtyard, inside the covered strip

The owner's answers to Phase 0 (9 Nov 2026) applied. Every core by its **outer faces** (RC walls 0.20, **ASSUMED**); its north wall is the y 6.5 wall rebuilt in RC (nothing taken from the women's majlis or the family living); its west face on x 4.30 keeps the gallery's west passage **1.52 m clear**. GF: the outlet (green) and the door (brown). FF: the passage widths (red). Roof: the head may grow north over the family living's roof (D7: ≥ the shaft, a funnel). 1 m = 0.415 cm.

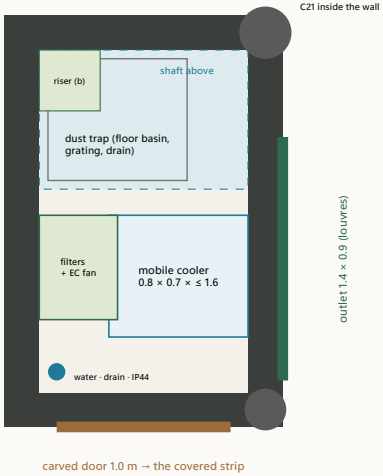


Variant	Shaft clear	Breeze area	Foot chamber	Outlet (free)	Door	Fits (cooler / filters / dust trap)	What is lost	Breeze (L1 V9.3 = 1)
V-a · corner core (west half of the strip)	0.4 × 0.8 = 0.32 m²	0.2 m²	0.4 × 0.8 = 0.32 m²	0.7 × 0.9 (0.38 m²)	0.5 m	no: 0.40 m clear wide, no cooler, no door ≥ 0.9	strip 0.9 m² · gallery 0.9 m² · riwaq glass 0.8 m solid · W1-1 untouched (a 0.7 m dead pocket beside the core)	**1.2** (≈ 0.53 m³/s)
V-b · full-strip core	1.2 × 0.8 = 0.96 m²	0.84 m²	1.2 × 0.8 = 0.96 m²	0.8 × 0.9 (0.43 m²)	1 m	tight: the cooler OR the filter cabinet, no standing room, the door must slide	strip 1.7 m² · gallery 1.7 m² · riwaq glass 0.8 m solid · W1-1 covered (Phase 3)	**2.7** (≈ 1.12 m³/s)
V-c · full-strip core, the foot chamber in the first arcade bay	1.2 × 0.8 = 0.96 m²	0.84 m²	1.2 × 1.97 = 2.36 m²	1.4 × 0.9 (0.76 m²)	1 m	yes: the cooler at the outlet, the filter cabinet + EC fan, the dust trap, a standing space at the door	strip 3.5 m² · gallery 1.7 m² · riwaq glass 2.1 m solid · W1-1 covered (Phase 3)	**3.5** (≈ 1.48 m³/s)

Finishes (owner, 9 Nov 2026): render and paint, no stone. The L5 core and its head are designed in **render and paint**: a light sand render; the projections (the cornice band below the head, the crowning Hijazi cornice, the arch surrounds of the head's openings and of the courtyard outlet) in the optional calm honey tone. Every projection, arch and cornice is **formed in the RC core and block** (not applied), with a drip and a fall on its top; plaster mesh over every block / concrete junction before rendering. No stone anywhere on the tower (this replaces the "honey stone" of `studies/malqaf_tower.typ` and of the V9.3 tower). Phase 5 designs the head on this basis.

Breeze: the air study's loss chain referred to its 0.10 m² section, with the same head pressure for all (ΔCp 0.8, the summer-evening mean wind 21 km/h): L1 (V9.3, the channel at full width) 0.42 m³/s. The L5 head's pressure is **not yet** the same (§ 1.3). Riser = the filtered path's Ø315 duct compartment (0.12 m²) inside the core. All **ASSUMED**.

Phase 1 · the foot chamber (V-c), the outlet, the door · the passage limit · the wind shadows



V-c · the GF foot chamber · 1 m = 2.3 cm

1.1 · The passage limit sets the shaft (a finding)

- At the first floor the gallery's north arm (x 2.75–4.25, beside the stair) turns into the strip (x 4.25–5.80) **between the core's south face and bedroom 2's bath** (its north-east corner at (4.35, 9.10)). For ≥ 1.5 m there the core's south face must stay at **y ≤ 7.6**: the shaft is at most **1.2 × 0.8 m clear (0.96 m²)**, the same order as L1. It is enough aerodynamically (its loss is ≈ 2 % of the chain: the head and the outlet govern), but it is **not** the "large" shaft of D7. **Options:** accept a 1.2 m turn (the shaft 1.2 × 1.1 = 1.32 m²), or move bath 2's north-east corner (a room change).
- **The foot chamber is free of that limit:** under the gallery floor it may run south; V-c ends it at the **first arcade column (y 8.667)**, so the outlet fills **the first arch** of the courtyard arcade, facing the seating along the colonnade line (no channel, the route ≈ 1 m).

1.2 · The foot chamber (V-c), the outlet, the door

- **Clear 1.20 × 1.97 m (2.36 m²), ≈ 3.1 m high (to the FF slab) and open above into the shaft.** The dust trap: a 0.8 × 0.7 floor basin under the shaft (a lift-out stainless grating, a 0.25 m deep sump, a drain to the courtyard drainage); the filtered path's riser in the north-west corner, its filter cabinet + EC fan (0.45 × 0.60) at the west wall; the cooler's place at the outlet; a water point with a float valve, a floor drain, a dedicated circuit and IP44 sockets.
- **The cooler (D9):** the common portable desert coolers sold in the Kingdom run from ≈ 45 L tanks (≈ 0.4 × 0.35 × 1.0 m) to 90–150 L (≈ 0.6–0.8 × 0.5–0.7 × 1.3–1.6 m); the chamber is laid out for the larger envelope **0.8 × 0.7 × 1.6 m** (**ASSUMED**: the owner's shortlist to confirm), rolled in through the door, its outlet against the grille.
- **The outlet: 1.4 × 0.9 m** in the first arch, sill +0.10, adjustable timber louvres or a pierced precast-concrete grille, rendered and painted (≈ 60 % free, 0.76 m²) aiming the air along the seating; a motorised insulated damper behind it (Phase 4).
- **The door: 1.0 m clear, full height (≈ 2.4 m)**, a carved timber door on the chamber's south face, opening onto the covered strip (the strip south of the core, 1.55 × 4.2 m, stays the riwaq's courtyard edge).

1.3 · The wind shadows (first estimate; Phase 5 confirms)

- **The old house** (one storey, ≈ 4–5 m, the owner): for a due-west wind its east face is **13 m upwind** of L5 (12.6 m for WSW; **nothing upwind for WNW–NW**). Its recirculating wake is ≈ R = 7.6 m long and ≈ 1.5–2 H (≈ 7–9 m) high near it: **L5 is beyond it**, and a head above ≈ +10 is above its decaying wake. ✓
- ⚠ **CAUTION · the new house's own roof matters more:** L5 stands 5.1 m behind the west facade, where the flow separating over the parapet and the attic is at its highest: R = 12.5 m, the recirculation bubble ≈ 2.7 m high at 6.2 m from the west edge (L1 sits near its tail, ≈ 11.2 m). With the L1 head's openings (+10.35 … +11.73) the L5 west face would breathe partly **inside** the bubble (top ≈ +11.42). **Phase 5 to set the L5 openings from ≈ +11.0 up to ≈ +12.6, the crown ≈ +13.2 (≈ 4.5 m above the attic)**, or accept a weaker west face in a pure west wind (the north face still catches NW).

1.4 · The on-site weather station (the owner's question)

- **Where:** on the build land at the L5 spot or as close as possible (≈ 6 m east of the M line, ≈ 14 m south of the gate), in the open, **not** on the old house's roof (its own disturbance).
- **Height:** the anemometer **≥ 6 m above the ground, ideally 10 m** (the airport's standard, and the tower head's ≈ +12 m), on a guyed telescopic mast; a second, cheap sensor at ≈ 4 m would show the old house's shadow.
- **What:** an ultrasonic anemometer (no moving parts) or a cup + vane, temperature / humidity, a PM2.5 sensor if possible; **1-minute averages**, logged; **≥ 4–6 weeks**, the evenings with a west wind included. We compare it with the airport for the same days.

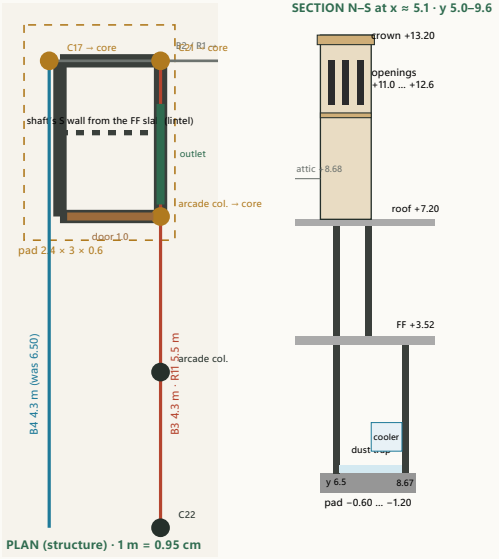
Open questions for the owner (Phase 1): (1) V-c (recommended) with the shaft limited to 0.96 m² by the gallery's 1.5 m turn, or accept a 1.2 m turn for a 1.32 m² shaft? (2) The riwaq's first glass panel (y 6.8–8.9, 2.1 m) becomes solid in V-c (the glass door and the 2.0 m south panel stay): OK? (3) The carved door on the chamber's south face into the covered strip: OK? (4) The Phase 0 answer had blanks: the farm's distance and direction from the airport, the trees or hills to the west; and whether to add the winter late mornings (Dec–Jan 10:00–13:00 average ≈ 20–23 °C) to the seating hours.

ملخص المرحلة 1: قُورنت ثلاثة بدائل للملقف في الركن الشمالي الغربي للفناء. الممر العلوي يفرض حدا: ليبقى عرض الالتفاف 1.5 م عند حمام غرفة النوم 2 لا يتجاوز المجرى في الدور الأول 1.2 × 0.8 م صافيا (0.96 م²)، وهذا يكفي هوائيا لأن الخسارة الكبرى في الرأس والفتحة. **التوصية: البديل (ج):** غرفة القاعدة في الدور الأرضي تمتد إلى أول عمود في الرواق المقوّس (1.2 × 1.97 م صافيا) وفيها مصيدة الغبار وخزانة المرشحات والمروحة ومكان المبرد الصحراوي المتنقل، وفتحة الهواء 1.4 × 0.9 م داخل أول قوس مقابل الجلسة مباشرة، وباب خشبي منحوت بعرض 1 م. النسيم المقدّر نحو 3.5 أضعاف الموقع الحالي. البيت القديم على بعد 13 م غربا فلا يحجب الرأس، لكن سقف البيت الجديد نفسه يحدث دوامة فوقه فيجب رفع فتحات الرأس إلى نحو +11 حتى +12.6 م (المرحلة 5). محطة الطقس: على أرض البناء في موضع البرج، بارتفاع 6 م على الأقل والأفضل 10 م، لمدة 4–6 أسابيع.

Phase 1 checks **passed** (5) · sources: data/rooms_gf.csv / rooms_ff.csv (V9.3), studies/structure.typ (C17, C21, B3, B4, R11), 3d/character.yaml (the arcade columns at y 8.667 / 10.833), data/deed_sketch.csv (the old house), ASHRAE Handbook — Fundamentals 2021, ch. 24 (Airflow around buildings: R, Hc, Xc, Lc); cooler sizes from Saudi and Gulf retail listings (Carrefour KSA 90 L, Aftron / Alhayyat / ALM 150 L).

Phase 2 · structure · the L5 core as an independent RC core, the fourth support of the west projection

The owner's Phase 1 answers applied: **V-c**, the shaft 1.2 × 0.8 m clear (the gallery's turn kept at 1.5 m); the riwaq's first glass panel solid; the carved door on the chamber's south face; **the winter late mornings added** (Dec–Jan 10:00–13:00: ≈ 61 % pleasant, ≈ 113 h a year more, ≈ 1812 h in all); **no stone** (render and paint). Every figure **ASSUMED** (no soil report, no SBC 301 site data): concept values for the structural engineer.



2.1 · The core

- **An independent RC core** (walls 0.20) from its own pad (−0.60) to the head: the shaft on the wall centre lines x 4.4–5.8, y 6.5–7.5 (outer 1.6 × 1.2), the foot chamber's walls to the arcade column line y 8.667 at the ground floor. Its north wall **is** the y 6.5 wall rebuilt in RC (the majlis's face unchanged); the riwaq's glazing on x 4.25 is not load-bearing and stays a glazing line south of the chamber.
- **Absorbed: C17** (4.25, 6.5) into the core's north-west corner, **C21** (5.8, 6.5) into its north-east corner, the decorative **arcade column at y 8.667** into the chamber's south-east corner (it becomes structural). **Kept:** C22, C23 and the second arcade column (y 10.833).
- **One pad** 2.4 × 3 × 0.6 m under the core and the chamber (it replaces three pads), with a 0.35 m recess for the dust trap's sump.

2.2 · The fourth support (the owner's question)

1. **B4** (x 4.25): its first span **6.50 → 4.3 m** (cast with the chamber's west wall).
2. **B3** (the edge, x 5.80): **6.50 → 4.3 m** on the chamber's east wall; **R11** (the roof edge) 6.50 → 5.5 m on the shaft.
3. **Reduced:** the first-span moments ≈ 56 % less, the deflections ≈ 80 % less: B3 and B4 could drop from 0.25 × 0.50 to ≈ 0.25 × 0.40 there (B3's upturn stays: it is W1's sill); C21 is no longer a separate column; the core also stiffens the west projection sideways (the slender C22 / C23 sway less).
4. **C21 and C17** disappear as separate members; B2's and R1's first span (C17 → C21, 1.55 m) becomes the core's north wall.

Beam	Line · how	Span was → now (m)	q kN/m	On the core kN
B3	x 5.80, FF · bears on the chamber's east wall to the arcade column line	6.5 → 4.33	12	52
B4	x 4.25, FF, drop · cast with the chamber's west wall (it overlaps the wall's outer 0.075 m: a local widening or a nib)	6.5 → 4.33	25	108.3
R11	x 5.80, roof · bears on the shaft's east wall	6.5 → 5.5	10	37.5
B2 / R1	y 6.5, FF / roof · their first span (C17 → C21) is the core's north wall	1.55 → –	10	14

Vertical loads (service, ASSUMED)	**kN**
the shaft's walls (−0.60 → +7.20)	162.5
the foot chamber's walls (less the outlet and the door)	58.6
the head above the roof (1.6 × 1.6, to +13.20, the cornices, the top slab)	162.2
the beams and slabs above (the table on the left)	211.8
total on the pad	**595.1**
the pad's bearing (with its weight) vs 150 kPa assumed	97.7 kPa

2.3 · The tower above the roof: wind and earthquake

- **The head** is a cantilever 5.9 m above the roof finish, 1.6 × 1.6 m (the shaft 1.6 × 1.2 plus a 0.4 m funnel overhang north on the head's base slab, D7), its self-weight ≈ 162.2 kN.
- **Wind** (q 1.1 kPa, Cf 1.3, **ASSUMED** SBC 301 for Taif to the engineer): 13.5 kN, **39.8 kN·m** at the roof.
- **Earthquake** as a rooftop appendage (SBC 301 / ASCE 7 § 13.3, ap 2.5, Rp 2.5): with Sds 0.15 / 0.25 / 0.35 (**ASSUMED** range for Taif) Fp = 29.2 / 48.7 / 68.1 kN, **86.1 / 143.5 / 201 kN·m** at the roof: **the earthquake governs**, not the wind. The RC tube takes it easily (≈ 0.31 MPa, the openings ignored): the head is **an RC frame** (four corner piers ≥ 0.30 × 0.30, a ring beam at the band, one at the crown) with the slender ribs between the openings as **non-structural** block or precast infill, rendered.
- **Connections:** the FF slab and the roof slab cast into the core (starter bars), so the core is tied at both levels; the shaft's openings in the slabs are framed by its own walls (no trimmers); the head's base slab cantilevers 0.4 m north over the family living's roof.

2.4 · Openings in the core: lintels and limits

- **Ground floor:** the outlet 1.4 × 0.9 in the east wall (sill +0.10; a lintel at +1.00, the wall above deep); the south wall is the door: a 1.0 m door in 1.2 m clear, i.e. the wall becomes a 1.4 m lintel at +2.40 over it on the east and west walls; the shaft's south wall starts at the FF slab on a 1.2 m lintel across the chamber.
- **First floor and above** (Phase 3 / 4): the gallery window into the shaft and the filtered-air grilles ≤ 50 % of a wall's length, ≥ 0.25 m from the corners, lintels ≥ 0.30 deep.

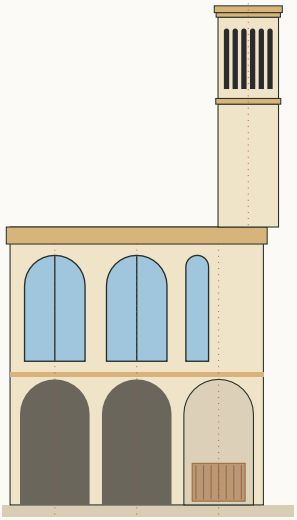
Open questions for the owner (Phase 2): (1) The head 1.6 × 1.6 with a 0.4 m funnel overhang north, or the head = the core (1.6 × 1.2) with only the cornices projecting? (2) Agree that the head is an RC frame with block / precast ribs between its openings (rendered)? (3) The Phase 0 / 1 blanks are still open: the farm's distance and direction from the airport, what lies to the west. **For the engineer:** the soil report (q allowable), SBC 301 wind and Sds for the site, B3 / B4 sections after the shorter spans, the pad and the slabs' ties to the core.

ملخص المرحلة 2: الملقف نواة خرسانية مستقلة بجدران 20 سم على قاعدة واحدة (2.4 × 3.0 م) من الأساس إلى الرأس، وجدارها الشمالي هو جدار المجلس نفسه بعد صبه خرسانة. تبتلع النواة العمودين C17 وC21 وعمود الرواق الأول (فيصير إنشائيا)، فتصبح الدعامة الرابعة للبروز الغربي: يقصر بحر الجسرين B3 وB4 من 6.5 م إلى نحو 4.3 م (العزوم أقل بنحو 55٪ والترخيم بنحو 80٪)، ويمكن تخفيف مقطعيهما، وتزيد صلابة البروز جانبيا. الحمل الكلي نحو 595 كيلونيوتن، والضغط على التربة نحو 98 كيلوباسكال (مفترض). الرأس فوق السطح يحكمه الزلزال لا الرياح، ويُبنى إطارا خرسانيا بأعمدة في أركانه والأضلاع بين الفتحات حشوة غير إنشائية مليّسة. كل الأرقام تقديرية للمهندس الإنشائي.

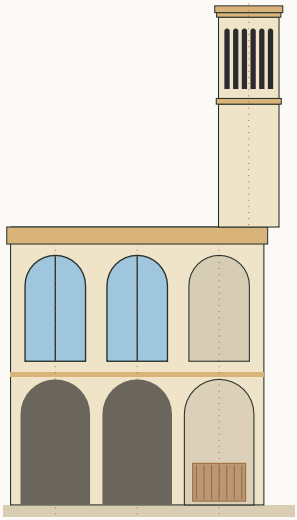
Phase 2 checks **passed** (3) · sources: studies/structure.typ (C17, C21, C22, B2, B3, B4, R1, R11, the projection slab), 3d/character.yaml (the arcade columns), SBC 301 / ASCE 7-16 § 13.3 (the appendage formula), the climate script for the winter mornings. Concept only.

Phase 3 · the courtyard’s west face · the malqaf as the head of the colonnade · three options

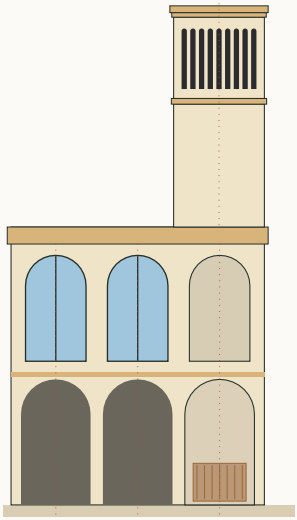
The owner’s Phase 2 answers applied: the head 1.6 × 1.6 with a 0.4 m funnel overhang north; an RC frame with corner piers and ring beams, the ribs in rendered block; **no stone and no marble anywhere, floors included** (marble-look porcelain inside, anti-slip R11 porcelain outside, recorded). Render and paint: a light sand render, the projections (the band, the cornices, the arch surrounds) in the calm honey tone. Red dotted: the bay axes (GF arches = FF windows) and the head’s axis. 1 m = 0.5 cm.



F1 · west face from the seating (N →)



F2 · west face from the seating (N →)



F3 · west face from the seating (N →)

3.1 · The three options

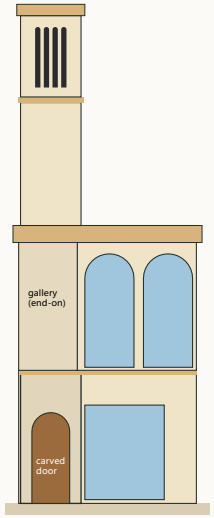
F1	the minimum: W1-1 replaced by a 0.6 m arched window beside the core (off the bay axis); the head as decided (north overhang) · the head’s axis −0.78 m from the first bay’s axis
F2	the whole first bay is the malqaf’s: the outlet in the first arch, a blind arched recess above it on the same axis; the head as decided (north overhang) · the head’s axis −0.78 m from the first bay’s axis
F3	as F2, and the head 1.6 × 2.4 m over the whole first bay (overhang 1.2 m south over the gallery roof, on R11 / R3): one vertical from the arch to the crown · the head’s axis 0.02 m from the first bay’s axis

- **Kept in all:** W1-2 and W1-3 (bays 2–3) and the three living-bay windows W1-4…6 on their axes; their sill / springing / crown logic (the sill proposed at **+3.80**, 2.80 m high, on the module); the GF arcade’s bays 2–3 open; the band at +3.40 and the main cornice in the honey tone.
- **The first bay** (y 6.5–8.667) becomes the malqaf’s: at the GF the outlet (1.4 × 1.0, timber louvres) fills the first arch, the rest of the arch a rendered infill of the core; at the FF the core covers the north 0.8 m of W1-1, so W1-1 goes: F1 puts a narrow 0.6 m window beside the core (off the bay axis: ⚠ D11); F2 and F3 make **a blind arched recess** (1.6 × 2.80, 0.10 deep, render) on the bay axis: the arch, the recess and the tower read as one.
- **The head:** F1 / F2 keep the decided north overhang: the head stands **0.78 m north of the first bay’s axis**, half of it beyond the colonnade’s end; F3 lengthens the head south over the gallery roof (1.6 × 2.4 m, its walls on R11 and R3) so it rises **exactly on the first bay’s axis**: the tower begins the colonnade.

Recommendation: F3

The arch with the outlet, the blind arch and the head on one vertical axis make the malqaf **the head of the row** without touching bays 2–3 and the living bay; the head grows to 3.84 m² (a bigger funnel, D7). It changes one Phase 2 answer: the funnel overhang goes **south over the gallery roof (1.2 m)** instead of north (0.4 m). If the owner keeps the north overhang: F2.

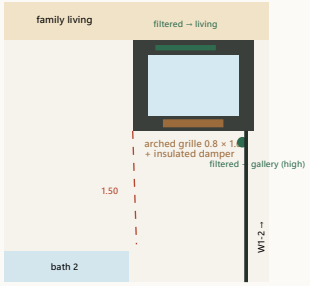
Phase 3 · from the courtyard looking north · from the riwaq · the gallery's grille into the shaft · the filtered-air grilles · D11



looking NORTH from the seating (W ←)



the RIWAQ looking east (N ←)



FF plan of the core · 1 m = 1 cm

3.2 · The gallery's window into the shaft (path a, clear days)

- On the shaft's **south face** at the FF (facing the gallery's turn): an **arched carved-timber grille 0.8 × 1.6** (sill +4.20, springing +5.40, a semicircular head; the W1 language in small) with an **insulated, low-leakage motorised damper** behind it in the wall: nothing swings into the 1.5 m passage or into the 0.8 m shaft. Open only in the **Clear day** mode (Phase 4); closed it reads as a carved panel.

3.3 · The filtered-air grilles (path b)

- From the fan in the foot chamber up the sealed riser: a high linear grille (0.8 × 0.2) through the core's **north wall into the family living** (+6.4, acoustically lined), one on its **south face into the gallery** (+6.4, above the timber grille); at the GF high grilles into the **riwaq** (the core's west face, +2.7) and the **women's majlis** (its north wall, +2.7). D10 kept: no bedroom touches the core.

3.4 · What else changes

- ⚠ **CAUTION** · the women's majlis loses its small courtyard glass panel at x 5.1–5.6 (0.5 m, inside the core's north wall); its main courtyard glass x 6.0–8.1 stays.
- The riwaq's first panel (y 6.8–8.9) becomes the core's rendered west face: a built-in timber **bench** along it (the riwaq's resting place) and the high filtered-air grille.
- The carved door (1.0 × 2.4, a semicircular head) faces the covered strip, seen from the seating beside the first arch.

3.5 · D11 compatibility (not designed here)

- **On the module:** every new size above (checks below); W1's sill at +3.80 (not +3.82) makes it 2.80 m high.
- ⚠ **CAUTION** · **D11 bands around each W1** (1.6 + 2 × 0.30 = 2.20 m) are wider than the arcade's bay (2.1667 m): the bands of neighbouring windows would overlap. The facade study should read D11 on this face as **shared pilaster bands on the column axes** (0.30, 1.87 m clear between them, W1 1.6 inside): the GF columns and the FF pilasters on the same axes.
- ⚠ **CAUTION** · the arcade's bays (6.50 / 3 = 2.1667 m) are **not** on the 20 cm module (the structure's grid): keep them, or adjust at the facade study.

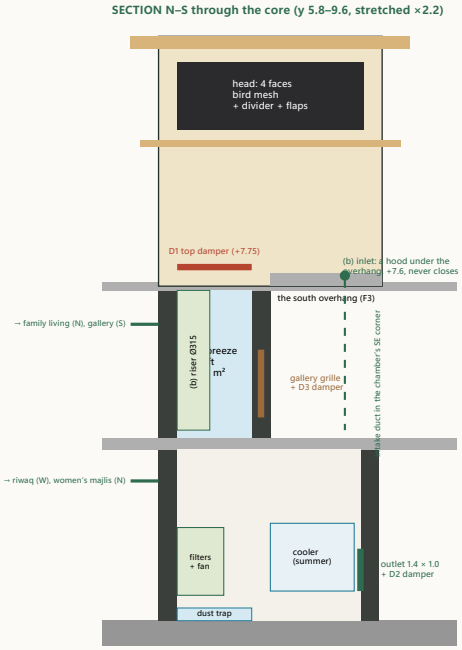
Open questions for the owner (Phase 3): (1) **F3** (the head 1.6 × 2.4 over the first bay, its overhang south over the gallery roof, on the colonnade's axis) or **F2** (keep the decided 1.6 × 1.6 with the north overhang)? (2) The gallery's "window" into the shaft as a carved arched grille with an insulated damper behind it (not an opening casement): OK? (3) W1's sill at +3.80 (on the module): OK? (4) The bench along the riwaq's new solid wall: OK? (5) The farm's location and what lies to the west are still blank.

ملخص المرحلة 3: يصبح الباكية الأولى من الرواق المقوّس باكية الملقف: في الأرضي فتحة الهواء بشيش خشبي داخل القوس الأول، وفي العلوي قوس أعمى غائر بنفس محور القوس (بدل النافذة الأولى التي تغطيها النواة)، وتبقى النوافذ الأخرى على محاورها فوق أقواس الأرضي. **التوصية F3:** يمتد رأس البرج جنوبا فوق سقف الممر (1.6 × 2.4 م) ليقوم تماما على محور الباكية الأولى، فيبدأ البرج صف الأعمدة بخط رأسي واحد من القوس إلى التاج؛ وإن بقي البروز شمالا كما تقرر فالخيار F2. نافذة الممر إلى المجرى شبكة خشبية منحوتة مقوّسة خلفها خانق معزول، ومخارج الهواء المرشح عالية إلى الممر وجلسة العائلة والرواق والمجلس. يفقد المجلس لوح زجاج صغيرا (0.5 م)، وفي الرواق كنبة خشبية على الجدار الجديد. ملاحظة لدراسة الواجهات: أشرطة D11 حول كل نافذة أعرض من الباكية، فالأفضل أن تكون أكتافا مشتركة على محاور الأعمدة.

Phase 3 checks **passed** (3) · sources: 3d/params.yaml (W1, W2), 3d/character.yaml (the arcade), presentation.typ (the courtyard glass), studies/roof.typ (the cornice, the band). Render and paint only (owner, 9 Nov 2026).

Phase 4 · the air system · two paths after the head · dampers · positive pressure · exhausts

The owner's Phase 3 answers applied: **F3** (the head 1.6 × 2.4 over the first bay, the overhang south over the gallery roof), the carved grille with its insulated damper, W1's sill +3.80, the bench, the majlis's 0.5 m panel lost, D11 as shared pilasters later; the farm's location from the on-site station. Every figure **ASSUMED**: the MEP engineer designs and sizes it.



4.1 · The two paths after the head

- **The head** (Phase 5): four faces of slender openings +11 … +12.6 behind a 20 mm **bird / leaf mesh**; inside, the + divider and the four **back-draught flaps** (the windward quarter feeds, the leeward quarters shut).
- **(a) The breeze path:** the head → **D1**, the top damper at the shaft's head (+7.75, 1.2 × 0.8) → the shaft (0.84 m²) → the **dust trap** (a 0.8 × 0.7 × 0.3 basin under the shaft, a lift-out grating, a drain) → the foot chamber → the **courtyard outlet** 1.4 × 1.0 with its louvres and **D2** behind them → the seating. **Optional:** the gallery's carved grille with **D3** (Clear day). **No filter:** mesh only, by design (D3 of the owner).
- **(b) The filtered path:** its **inlet never closes:** a downward-facing hood (0.6 × 0.6, a bird mesh) **under the head's south overhang** (+7.6; rain- and dust-shadowed, 3.4 m below the breeze openings), a Ø315 duct down the chamber's south-east corner → the **filter cabinet** (ISO Coarse 60 % (G4), washable synthetic, 500 × 500 × 48 → ISO ePM1 ≥ 50 % (F7 class), 500 × 500 × 292 bag or compact; ePM2.5 ≥ 65 % if the PM2.5 log asks for more) → the **EC fan** (750 m³/h at 250 Pa, ≈ 160 W, ≈ 38 dB(A) in the chamber) → a sealed Ø250 riser in the shaft's west side → high grilles into the **family living** and the **gallery** (+6.4), and at the GF the **riwaq** and the **women's majlis** (+2.7); the other GF rooms through the riwaq (door undercuts / transfer grilles, as the house's air moves anyway). **The kitchen hood keeps its own interlocked make-up inlet** (≈ 300 m³/h, `studies/air_system.typ`).

4.2 · The dampers (all motorised, insulated blades, low-leakage class 2 / 3, spring-return: they CLOSE on power loss)

- **D1 top damper:** the main breeze valve; closes in Dust, Cold, at night if wanted; half in High wind. **D2 outlet damper:** the second seal of the courtyard outlet (the chamber then holds the cooler and the filters at room conditions). **D3 gallery damper:** the only breeze opening **into the house**: open only on a Clear day; insulated, gasketed, with an end-switch. **D4 chamber vent** (a small Ø160 damper to the outlet) lets the cooler's moist air out when the chamber is shut. The hood's make-up damper stays as designed.

4.3 · Positive pressure and the hood

- The fan supplies 325 m³/h (normal) / 704 (boost) into a **tight envelope (D5)**; the extracts (the bathrooms, ≈ 150–250 m³/h continuous, the dryer) are smaller: the house runs at ≈ **+5–10 Pa** (the dP sensor trims the fan). Every door is undercut 15 mm or has a transfer grille so the pressure is one.
- ⚠ **CAUTION** · the kitchen hood (600–1 000 m³/h) exceeds the supply fan: without its make-up inlet the house would go **negative** and pull dust through every crack and the malqaf's grille. The **interlocked make-up inlet** (filtered, motorised, opens with the hood) is a must; the fan may boost with it but does not replace it.

4.4 · Exhausts and plumes relative to the head (openings +11 … +12.6)

Source	Distance	Position
bathrooms' extract (the west stack's vent)	5.7 m	⚠ UPWIND in a W wind: the terminal taken ≥ 1.5 m above the roof (+8.8), 3.6 m from the head's west face: below the head's openings (+11.0) by 2.2 m; or re-routed in the attic to the south-east
kitchen hood outlet (east wall)	12.4 m	downwind, 7.5 m, 8.8 m below the openings
east stack's vent	8.7 m	downwind, 7.3 m
AC condensers Z2b (hot air)	5.8 m	east, 3.7 m in plan from the head's east face, 3.1 m below the openings; their hot plume rises leeward in a W wind
the plumbing's drain vents (the east stack, the family WC's stack)	8.7 m	south-west, 6.4 m; terminals ≥ 1.5 m above the roof
the LPG cabinet, the generator point (the yard)	14 m	downwind, ≥ 8 m, 11 m below

⚠ **CAUTION** · the bathrooms' vent in the west shaft is the one source **upwind** in the summer-evening west wind: its terminal must end ≥ 1.5 m above the roof (+8.8), 2.2 m below the head's openings and 3.6 m away, with a charcoal / membrane terminal; better: re-route it in the attic to the south-east strip (Phase 6 checklist).

Phase 4 · insulation and moisture · sensors · the control modes · maintenance · performance

4.5 · Insulation, acoustics, moisture

- **The core's walls facing the gallery and the family living:** 200 RC (Rw ≈ 55 dB) + 50 mm mineral wool and a 2 × 12.5 board lining **on the room side** (the breeze shaft stays smooth render inside); the fan's cabinet acoustically lined, on anti-vibration mounts; the filtered-air grilles with 0.6 m lined duct before them.
- **Thermal:** D1 and D3 insulated blades; the shaft's walls facing rooms with 50 mm insulation inside the lining (the shaft is outdoor air in winter).
- **The foot chamber (wet):** tanked to +0.30 and a fall to a **trapped gully** (a trap primer), the dust trap in GRP, the cooler on a drip tray, a float-valve water point, IP44 sockets at +1.2 on their own RCD circuit, a lockable isolator; a hygrometer-controlled **D4 vent** so the chamber never stays damp.

4.6 · Sensors (simple and robust)

- **Outdoor PM2.5** (a laser sensor in a weather hood on the head's north face, shaded) · **outdoor T / RH** (the same hood) · **the house dP** (indoor minus outdoor, a tube to a calm point on the roof) · **the filter dP** (across both filters: service at 200 Pa, clean ≈ 60) · **indoor CO2** (the gallery) · **the chamber's RH** (D4) · **the cooler's water level** (its float) · **the roof anemometer** (the weather station's sensor kept). A small PLC / home controller with the modes and **a manual override switch per damper** (the owner's rule).

4.8 · Maintenance (all from the carved door at ground level)

What	How	When
the dust trap	lift the grating, sweep / vacuum the basin, hose to the drain	monthly; weekly in Mar–Apr
the pre-filter	slide out, wash, dry, back	monthly (the dP light)
the fine filter	replace	≈ 6 months (dP 200 Pa)
the cooler	tank and pads cleaned; drained and parked for winter	weekly in summer; Oct
the dampers D1–D4	stroke test, seals	yearly (before spring)
the gully trap, the drain	fill / flush	monthly
the outlet louvres, the grilles	brush	3 months
the head's mesh and flaps	**the roof hatch only**: brush the mesh, free the flaps	yearly; after a dust storm

4.7 · The control modes (the thresholds of Phase 0, accepted by the owner; a manual override on every damper)

Mode	Switching (outdoor sensors)	D1 top	D2 outlet	D3 gallery	Cooler	Fan (b)	What happens
Normal	20 ≤ T ≤ 32 °C · PM2.5 ≤ 35 µg/m³ · wind ≤ 45 km/h	open	open	closed	off	normal 325 m³/h	the courtyard breeze; the house ventilated by the fan at a slight positive pressure
Clear day	18 ≤ T ≤ 30 · PM2.5 ≤ 15 · wind ≤ 30	open	open	**open**	off	minimum	the gallery and the family living get the natural breeze through the grille (path a into the house; the fan idles)
Dust	PM2.5 > 50 for 10 min (back to Normal below 25 for 30 min)	**closed**	**closed**	closed	off	**boost** 704 m³/h	every breeze opening shut; the filtered fan at boost keeps the house at +10 Pa: dust stays out
Cold	T < 18 °C (Nov–Mar: < 20 after sunset); frost: < 5 °C	closed	closed	closed	off	minimum (≈ 150)	the shaft and the chamber closed and insulated; the cooler drained in winter
Summer	T > 32 °C (Jun–Sep afternoons)	open	open	closed	**on**	normal	the cooler draws the malqaf air, blows ≈ 20–22 °C air through the outlet onto the seating; the gallery grille shut (humid air out of the house)
High wind	gusts > 45 km/h	**half**	open	closed	–	normal	less dust and noise; the flaps and the damper modulate
Hood	the kitchen hood on (any mode)	–	–	–	–	also the make-up	the hood's own interlocked make-up inlet opens (≈ 300 m³/h); the fan does not try to feed the hood

4.9 · Rough performance (ASSUMED, for the MEP engineer)

- **The breeze (Normal):** at the summer-evening mean wind (21 km/h at 10 m) ≈ **1.82 m³/s** through the outlet (≈ 2.2 m/s at the louvres, ≈ 0.5–0.8 m/s at the seating 2 m away: a clearly felt breeze); at 10 km/h ≈ 0.86 m³/s (gentle); in a calm the stack alone ≈ 0.15 m³/s. L1 (V9.3) at 21 km/h ≈ 0.42 m³/s: **≈ 4.3 ×** (Phase 1 said 3.5 with the smaller head; F3's head is larger).
- **Summer (the cooler):** a 90–150 L portable cooler moves ≈ 0.8–1.4 m³/s and cools the dry afternoon air from ≈ 35 to ≈ **20–22 °C** (Phase 0: RH 12–20 %); its water ≈ 15–25 L/h; the jet along the seating ≈ 1 m/s. The courtyard is open to the sky: it cools **the people in the jet**, not the courtyard.
- **Winter:** the breeze closed; the fan at its minimum ≈ 150 m³/h costs ≈ **0.51 kW** at 10 K (≈ 1.11 kW at the normal rate): small; no heat recovery (D2 of the owner), so the fan runs at the minimum on cold nights and the CO2 sensor raises it when the house is full.
- **Dust:** the fan at boost with the fine filter (≥ 50 % of PM1, ≈ 80 % of PM2.5) holds the indoor PM2.5 at roughly a quarter of the outdoor level during a 150 µg/m³ event (the house tight, D5).

Open questions for the owner (Phase 4): (1) The filtered path's inlet under the head's south overhang (never closes), the filters and the fan in the foot chamber: OK, or the fan in the west shaft's utility room instead (quieter for the courtyard, a longer duct)? (2) The gallery grille (D3) open only on Clear days: OK? (3) The bathrooms' vent: accept the raised terminal, or re-route it to the south-east? (4) A home controller the owner programs himself (the modes above), or a packaged BMS?

ملخص المرحلة 4: بعد رأس البرج مساران: (أ) مسار النسيم الكبير بلا مرشح (شبك طيور فقط): خانق علوي D1، ثم المجرى، ثم مصيدة الغبار في القاعدة، ثم فتحة الفناء 1.4 × 1.0 م بخانق D2 خلف الشيش، وشبكة الممر العلوي بخانق D3 لا تُفتح إلا في الأيام الصافية؛ (ب) المسار المرشح الصغير: مدخل لا يُغلق أبدا تحت بروز الرأس الجنوبي، ثم مرشح خشن قابل للغسل ومرشح ناعم ومروحة هادئة 750 م³/س في غرفة القاعدة، تدفع هواءً مرشحا إلى الممر وجلسة العائلة والرواق والمجلس فيبقى البيت تحت ضغط موجب. كل الخوانق معزولة وتُغلق عند انقطاع الكهرباء، ولكل منها مفتاح يدوي. تنبيه: شفاط المطبخ أكبر من المروحة فلا بد من مدخل الهواء التعويضي المرتبط به. تنبيه: تهوية الصرف في الشفت الغربي تقع في اتجاه الريح الغربية؛ تُرفع أو تُحوَّل إلى الجنوب الشرقي. الصيانة كلها من الباب المنحوت في الدور الأرضي، وفتحة السطح لشبك الرأس فقط مرة في السنة. النسيم المقدّر نحو 1.8 م³/ث في مساء صيفي عادي (نحو أربعة أضعاف الموقع الحالي بعد تكبير الرأس في F3)، والمبرد يخرج هواءً بنحو 20–22°م.

Phase 4 checks **passed** (4) · sources: studies/air_system.typ (the ventilation rates 325 / 704 m³/h, the hood's make-up, the loss chain), studies/plumbing.typ (the vents), studies/roof.typ (Z2b), ASHRAE 62.1 / 62.2-2022, ASHRAE 62.1 Table 5-1 (separations), ISO 16890 (the filter classes), EN 1751 (damper leakage classes). Concept only.